
Q1. What is the main purpose of virtualization?

- A. Increasing physical hardware size
- B. Creating physical machines
- C. Running multiple virtual environments on one machine
- D. Deleting operating systems

Answer: C

Explanation: Virtualization allows multiple virtual environments to run on a single physical machine.

Q2. Which of the following is a Type 1 hypervisor?

- A. VirtualBox
- B. VMware Workstation
- C. VMware ESXi
- D. Parallels

Answer: C

Explanation: Type 1 hypervisors run directly on hardware, and VMware ESXi is an example.

Q3. Type 2 hypervisors run on top of:

- A. BIOS
- B. Physical hardware directly
- C. Cloud servers
- D. A host operating system

Answer: D

Explanation: Type 2 hypervisors require a host OS like Windows or Linux to function.

Q4. What does a snapshot capture?

- A. The hardware state only
- B. Only user files
- C. The state of a VM at a point in time
- D. CPU processes only

Answer: C

Explanation: A snapshot preserves the complete VM state, including memory and disk.

Q5. What is cloning in virtualization?

- A. Deleting VM storage
- B. Making a backup
- C. Creating a copy of a VM
- D. Monitoring a VM

Answer: C

Explanation: Cloning creates an identical copy of a virtual machine.

Q6. Which of these is **not** a benefit of virtualization?

- A. Hardware underutilization
- B. Resource optimization
- C. Isolation
- D. Scalability

Answer: A

Explanation: Virtualization avoids hardware underutilization by sharing resources.

Q7. Which hypervisor type provides better performance?

- A. Type 2
- B. Type 3
- C. Type 1
- D. None

Answer: C

Explanation: Type 1 hypervisors have direct hardware access, hence better performance.

Q8. A VM template is:

- A. Used for logging
- B. A virtual CPU
- C. A read-only image used to create VMs
- D. A snapshot file

Answer: C

Explanation: Templates speed up the creation of new, standardized VMs.

Q9. OS virtualization is also known as:

- A. BIOS emulation
- B. Kernel-based virtualization
- C. Containerization
- D. Hardware virtualization

Answer: C

Explanation: Containerization (like Docker) is a form of OS-level virtualization.

Q10. Which of the following is used to detect node failure in clusters?

- A. DNS
- B. SSH
- C. Heartbeat
- D. Ping

Answer: C

Explanation: Heartbeat mechanism regularly checks node status to detect failures.

☒ Medium Level (30 Questions)

Q11. What is the key difference between full and para-virtualization?

- A. Para-virtualization doesn't require a hypervisor
- B. Full virtualization runs without OS modification
- C. Para-virtualization is slower
- D. Full virtualization uses less memory

Answer: B

Explanation: Para-virtualization requires a modified OS; full virtualization does not.

Q12. Which term refers to grouping servers for high availability?

- A. Hypervisor
- B. Cluster
- C. Container
- D. Snapshot

Answer: B

Explanation: Clustering involves multiple servers working together to ensure uptime.

Q13. Docker is an example of:

- A. Type 1 hypervisor
- B. Hardware virtualization
- C. Container-based OS virtualization
- D. BIOS virtualization

Answer: C

Explanation: Docker uses OS-level virtualization to run lightweight containers.

Q14. What is a linked clone?

- A. A clone with full disk copy
- B. A copy without network settings
- C. A VM that shares the original disk
- D. A snapshot

Answer: C

Explanation: Linked clones save storage by sharing base disk with the parent VM.

Q15. Which of the following is a benefit of clustering?

- A. Downtime during maintenance
- B. Reduced availability
- C. High availability
- D. Increased power usage

Answer: C

Explanation: Clustering ensures services remain available even if a node fails.

Q16. What is fencing in clustering?

- A. Creating virtual firewalls
- B. Isolating failed nodes
- C. Increasing storage
- D. Reducing memory

Answer: B

Explanation: Fencing isolates problematic nodes to protect cluster integrity.

Q17. What role does shared storage play in virtualization clusters?

- A. Stores ISO files
- B. Hosts VM disk files accessible to all nodes
- C. Logs cluster metrics
- D. Acts as swap memory

Answer: B

Explanation: Shared storage allows VMs to migrate between nodes.

Q18. What component is responsible for VM migration in clusters?

- A. Firewall
- B. Hypervisor
- C. Cluster management software
- D. BIOS

Answer: C

Explanation: Tools like vCenter or Proxmox manage VM migrations.

Q19. What is the purpose of heartbeat in clustering?

- A. Monitor VM load
- B. Update OS
- C. Detect node failure
- D. Manage backups

Answer: C

Explanation: Heartbeat helps identify if a node has gone down.

Q20. In para-virtualization, which component is aware of virtualization?

- A. BIOS
- B. Hypervisor
- C. Guest OS
- D. Network

Answer: C

Explanation: The guest OS is modified to interact with the hypervisor.

Q21. What is a major drawback of snapshots?

- A. They increase CPU load
- B. They replace backups
- C. They consume storage over time
- D. They change the base image

Answer: C

Explanation: Snapshots grow in size and can consume significant storage.

Q22. Templates are most useful for:

- A. Upgrading VMs
- B. Monitoring network
- C. Mass deployment of similar VMs
- D. Changing hypervisors

Answer: C

Explanation: Templates standardize and simplify VM deployment.

Q23. Which type of virtualization uses a hypervisor between hardware and OS?

- A. Application virtualization
- B. OS virtualization
- C. Type 1 virtualization
- D. None

Answer: C

Explanation: Type 1 hypervisors sit directly on hardware.

Q24. VMware Workstation is categorized under:

- A. Type 1
- B. Cloud-native
- C. Containerization
- D. Type 2

Answer: D

Explanation: VMware Workstation runs on top of a host OS.

Q25. Which one is most suitable for production environments?

- A. Type 1 Hypervisor
- B. Type 2 Hypervisor
- C. Container runtime

D. BIOS

Answer: A

Explanation: Type 1 offers better performance and security for production.

Q26. Which of the following best describes "vMotion" in VMware?

A. Live VM backup

B. VM replication

C. Live migration of VMs

D. VM suspension

Answer: C

Explanation: vMotion is VMware's live migration feature.

Q27. What does OS-level virtualization share among containers?

A. Disk

B. Kernel

C. BIOS

D. Drivers

Answer: B

Explanation: Containers share the host OS kernel.

Q28. Para-virtualization offers:

A. No hypervisor involvement

- B. Better performance with modified OS
- C. Slower execution
- D. Support only for Windows

Answer: B

Explanation: It improves efficiency by allowing the OS to communicate with the hypervisor.

Q29. Which clustering concept involves detecting failed nodes?

- A. Live migration
- B. Cloning
- C. Heartbeat
- D. vSwitching

Answer: C

Explanation: Heartbeat checks whether nodes are active.

Q30. In virtualization, a **template** is used for:

- A. Load balancing
- B. Backup
- C. Standard VM creation
- D. License management

Answer: C

Explanation: Templates simplify the deployment of standard VMs.

(Questions 31–50 continue below—let me know if you'd like me to proceed with the remaining **20 MCQs (10 medium + 10 hard)** now.)

You said:

yes

ChatGPT said:

Great! Here's the remaining 20 MCQs for "Introduction to Virtualization":

☒ **Medium Level (continued: Q31–Q40)**

Q31. What is the role of a hypervisor in virtualization?

- A. Hosts physical storage
- B. Allocates memory manually
- C. Manages VMs and allocates resources
- D. Builds hardware

Answer: C

Explanation: Hypervisors manage virtual machines and handle CPU, memory, disk, and network allocations.

Q32. Which one is **not** a clustering software?

- A. Kubernetes
- B. vSphere
- C. XenServer
- D. Docker

Answer: D

Explanation: Docker is a container runtime, not a clustering tool.

Q33. Which hypervisor supports both para- and full virtualization?

- A. VirtualBox
- B. Xen
- C. Hyper-V
- D. KVM

Answer: B

Explanation: Xen is capable of both para-virtualization and full virtualization.

Q34. What happens if the shared storage in a cluster fails?

- A. VMs are automatically backed up
- B. Live migration continues
- C. VM access and migration fail
- D. Cluster remains unaffected

Answer: C

Explanation: Shared storage is critical for VM operation and migration; its failure halts services.

Q35. Which container tool allows OS virtualization with isolated namespaces?

- A. Git
- B. Docker
- C. Jenkins
- D. NGINX

Answer: B

Explanation: Docker uses namespaces and cgroups for OS-level virtualization.

Q36. Why is consistency in cluster hardware important?

- A. To support backups
- B. To ensure predictable performance and compatibility
- C. For different VM templates
- D. For cloning

Answer: B

Explanation: Similar specs ensure load balancing and prevent incompatibility.

Q37. The primary function of a VM snapshot is to:

- A. Backup disk to external drive
- B. Rollback to a known VM state
- C. Encrypt the disk
- D. Clone the system

Answer: B

Explanation: Snapshots help revert to a stable state during upgrades or tests.

Q38. Cloning a VM differs from a snapshot because:

- A. It is not possible in virtual environments
- B. It includes logs
- C. It creates a new, independent VM
- D. It modifies the source VM

Answer: C

Explanation: A cloned VM is a full or linked copy and can run independently.

Q39. OS-level virtualization is limited by:

- A. Host kernel compatibility
- B. Lack of containers
- C. BIOS support
- D. Slow performance

Answer: A

Explanation: Containers share the host kernel; therefore, guests must be compatible.

Q40. Which of the following allows you to start a new container in Docker?

- A. docker create-image
- B. docker snap
- C. docker start
- D. docker run

Answer: D

Explanation: docker run is used to create and start a new container.

⚔ Hard Level (10 Questions)

Q41. Which component in clustering ensures isolated node cannot cause damage?

- A. Resource scheduler
- B. Hyperthreading
- C. Fencing
- D. RAID

Answer: C

Explanation: Fencing disables unresponsive or faulty nodes to protect cluster data.

Q42. Which hypervisor is built directly into the Linux kernel?

- A. Xen
- B. KVM
- C. VirtualBox
- D. Proxmox

Answer: B

Explanation: KVM (Kernel-based Virtual Machine) is part of the Linux kernel.

Q43. When using para-virtualization, what is replaced in the guest OS?

- A. Bootloader
- B. Network stack
- C. System calls
- D. Hardware drivers

Answer: D

Explanation: Special drivers allow the guest to interact efficiently with the hypervisor.

Q44. What is the correct sequence for a VM lifecycle?

- A. Create → Snapshot → Clone → Delete
- B. Create → Start → Snapshot → Stop → Delete
- C. Create → Clone → Export
- D. Import → Clone → Backup

Answer: B

Explanation: This reflects the standard sequence in managing VMs.

Q45. What command creates a read-only container image in Docker?

- A. docker template
- B. docker run --save
- C. docker commit
- D. docker clone

Answer: C

Explanation: docker commit saves a running container's state as an image.

Q46. Which of the following ensures VMs can be moved between cluster nodes without downtime?

- A. Static routing
- B. Network fencing
- C. Live migration
- D. NAT translation

Answer: C

Explanation: Live migration like VMware's vMotion keeps the VM active during the move.

Q47. In VMware, what must be consistent across nodes for VM migration?

- A. IP address
- B. CPU architecture and version
- C. Hostname
- D. NIC vendor

Answer: B

Explanation: Migration often requires matching CPU families for compatibility.

Q48. Which statement is TRUE regarding linked clones?

- A. They do not depend on a base VM
- B. They take more storage than full clones
- C. Deleting base VM breaks the linked clone
- D. They work on different hypervisors

Answer: C

Explanation: Linked clones rely on the parent VM disk; deleting it invalidates them.

Q49. What is a major concern when using many snapshots over time?

- A. Reduced boot time
- B. Increased CPU load
- C. Disk fragmentation and performance degradation
- D. Snapshot theft

Answer: C

Explanation: Multiple snapshots can slow VM performance due to disk chaining.

Q50. In cluster design, quorum is used to:

- A. Increase RAM
- B. Allocate CPU resources
- C. Determine the number of active nodes required for cluster operation
- D. Select the leader container

Answer: C

Explanation: Quorum ensures that a majority of nodes are functional before making decisions.